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Lab 3 :

```
data class Profile(val name: String, val title: String, val bio: String)
```

```
fun renderAvatar(width: Int, height: Int): String {  
    val sb = StringBuilder()  
    val innerWidth = width - 4  
    val label = "Avatar ${width}x${height}"  
    val paddedLabel = if (label.length > innerWidth) {  
        label.take(innerWidth)  
    } else {  
        val leftPad = (innerWidth - label.length) / 2  
        " ".repeat(leftPad) + label + " ".repeat(innerWidth - label.length - leftPad)  
    }  
  
    sb.append("| ").append(paddedLabel).append(" |\n")  
  
    for (i in 1 until height) {  
        sb.append("| ").append(" ".repeat(innerWidth)).append(" |\n")  
    }  
  
    return sb.toString()  
}
```

```
fun renderTextBlock(label: String, text: String, width: Int): String {
```

```
val innerWidth = width - 4

val content = "$label: $text"

val trimmed = if (content.length > innerWidth) {
    content.take(innerWidth)
} else {
    content
}

val padded = trimmed + " ".repeat(innerWidth - trimmed.length)

return "| $padded |\n"
}
```

```
fun renderProfile(profile: Profile, width: Int = 30, showAvatar: Boolean = true): String {

    val sb = StringBuilder()

    val border = "+${" ".repeat(width - 2)}+\n"

    sb.append(border)

    if (showAvatar) {
        sb.append(renderAvatar(width, 3))
    }

    sb.append(renderTextBlock("Name", profile.name, width))
    sb.append(renderTextBlock("Title", profile.title, width))
    sb.append(renderTextBlock("Bio", profile.bio, width))

    sb.append(border)

    return sb.toString()
}
```

```
}
```

```
fun main() {
```

```
    val p1 = Profile("Ada Lovelace", "Software Dev", "Early computing pioneer")
```

```
    val p2 = Profile("Alan Turing", "Mathematician", "Father of theoretical CS")
```

```
    val p3 = Profile("Grace Hopper", "Computer Scientist", "Developed first compiler")
```

```
    println(renderProfile(p1, 30, true))
```

```
    println(renderProfile(p2, 30, false))
```

```
    println(renderProfile(p3, 30, true))
```

```
}
```